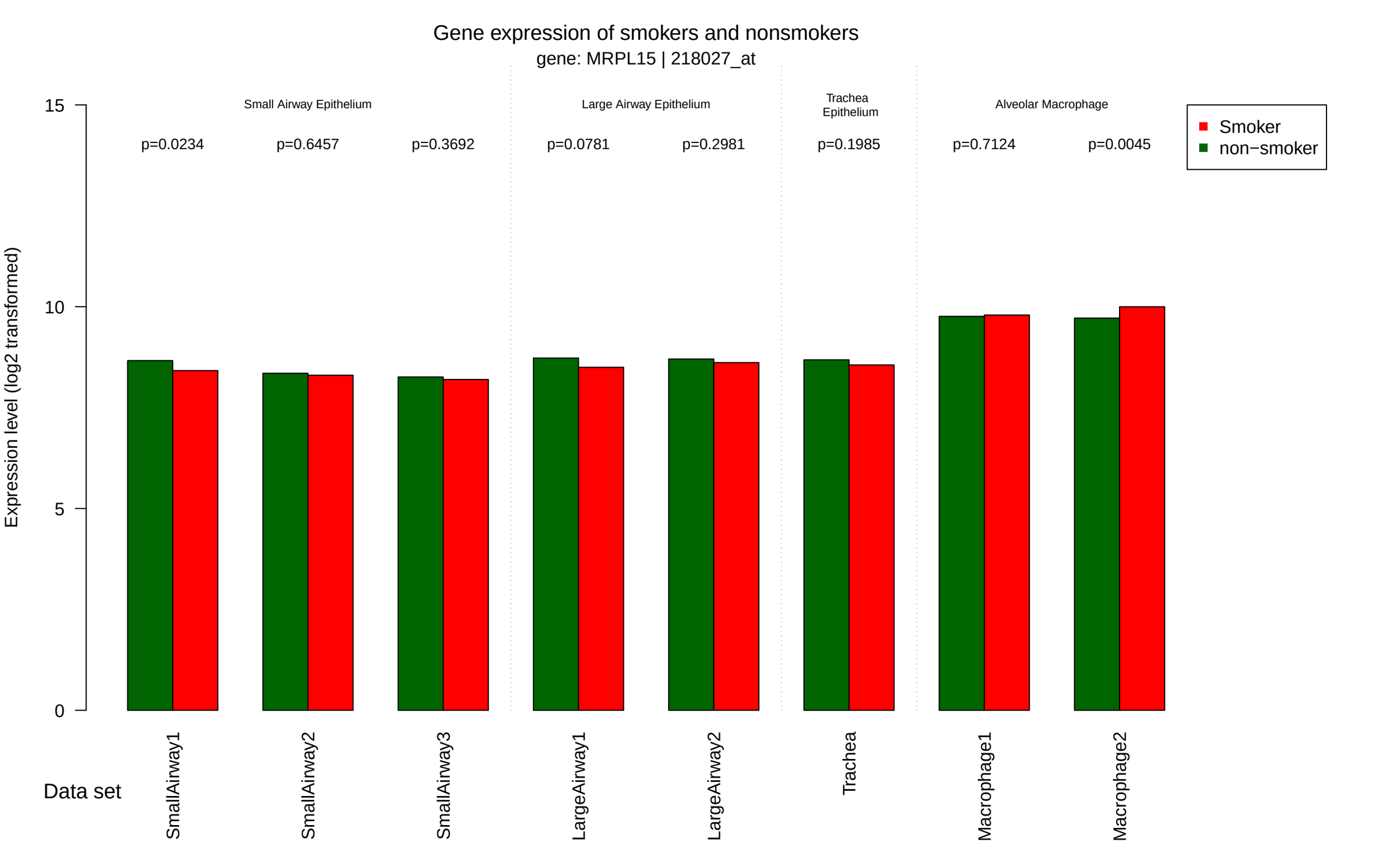
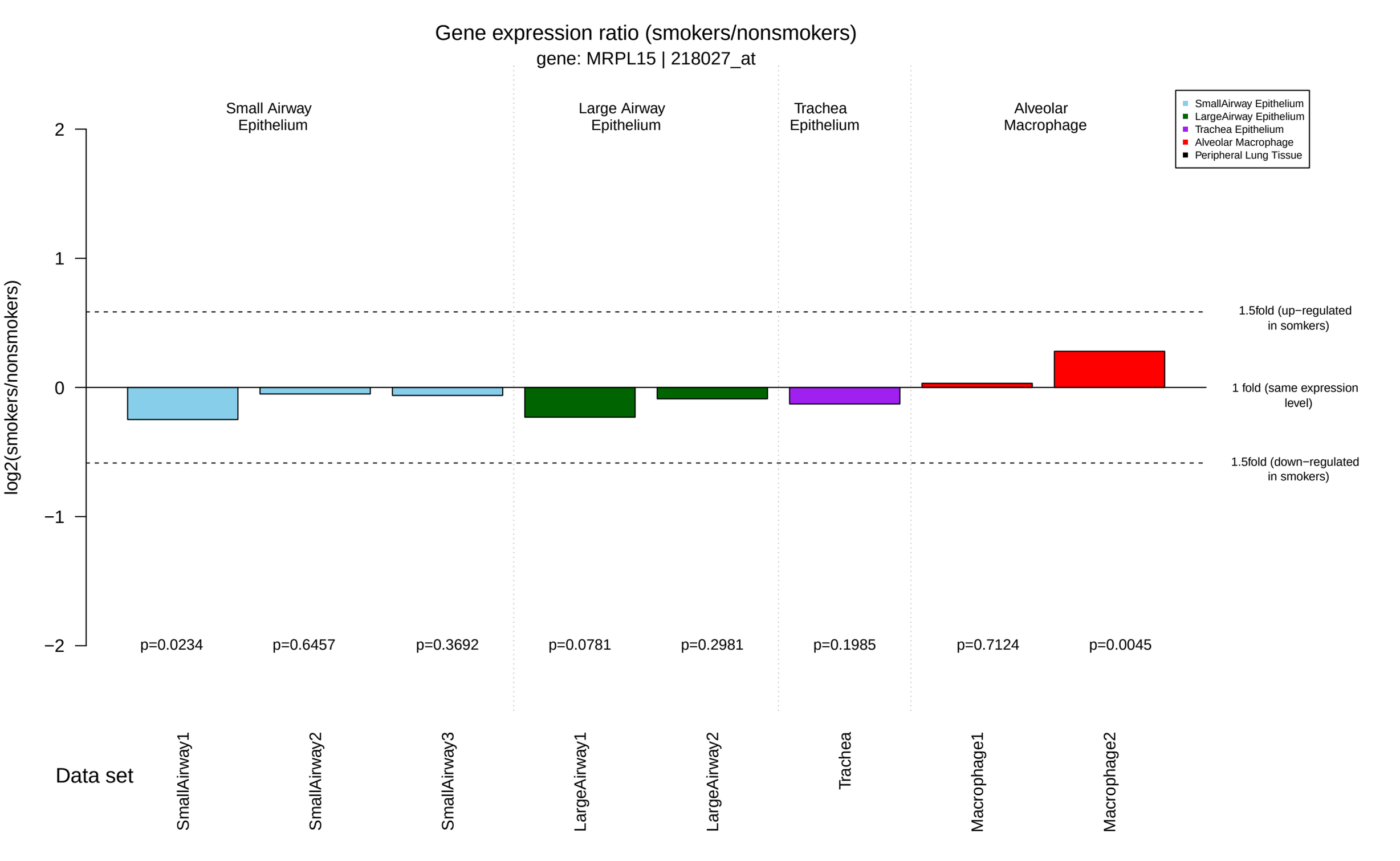
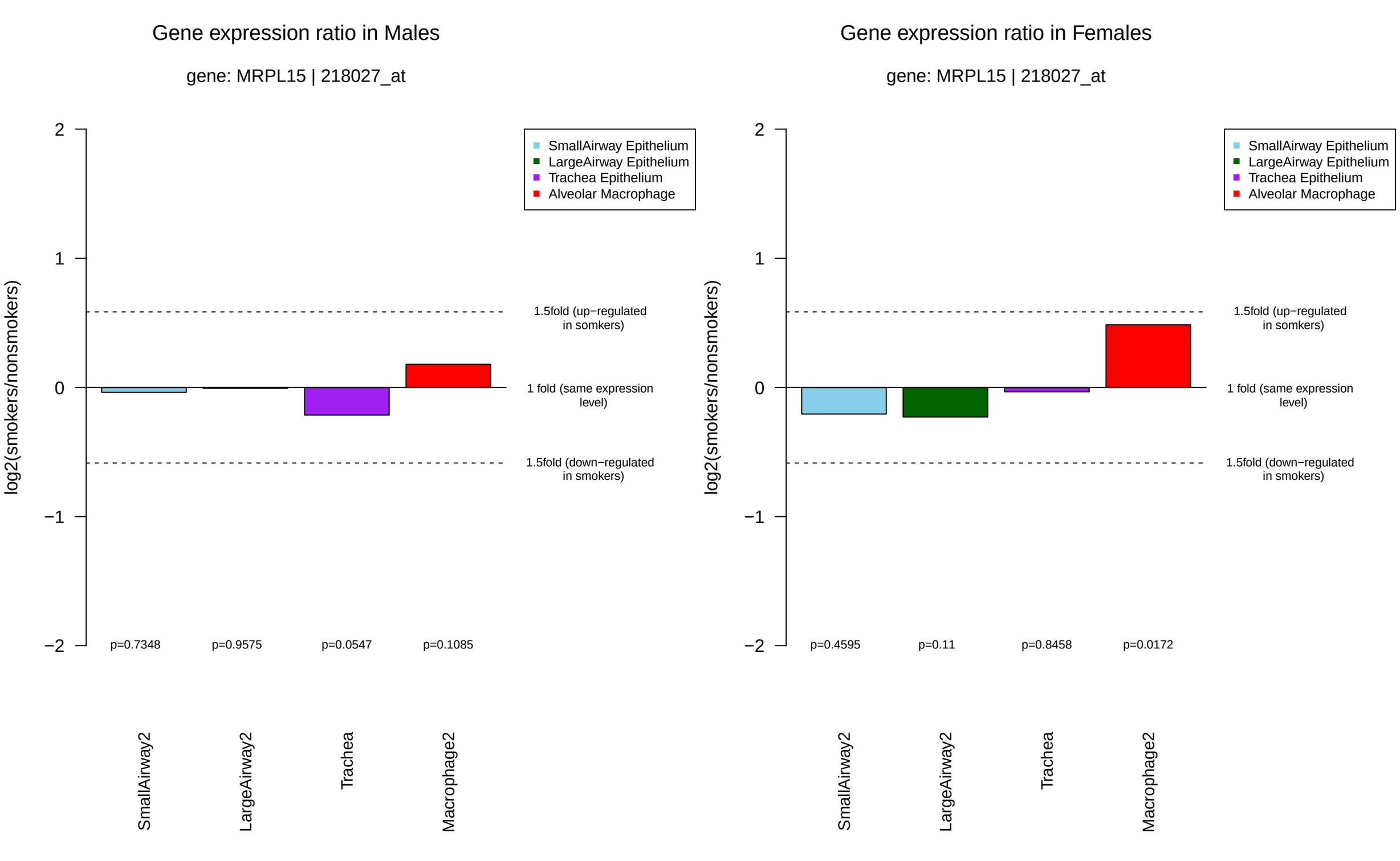
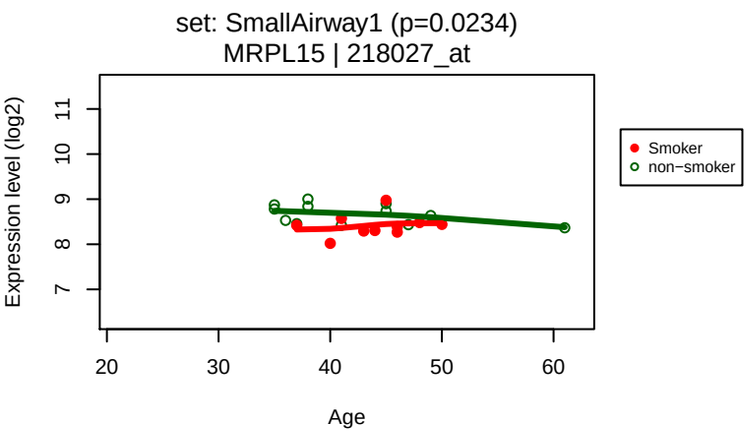
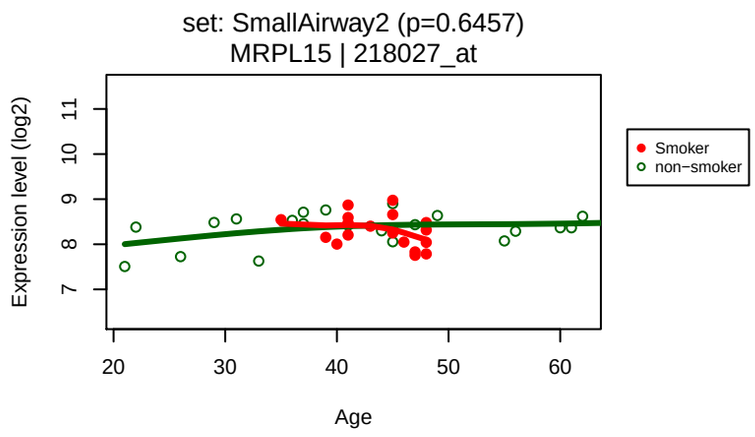
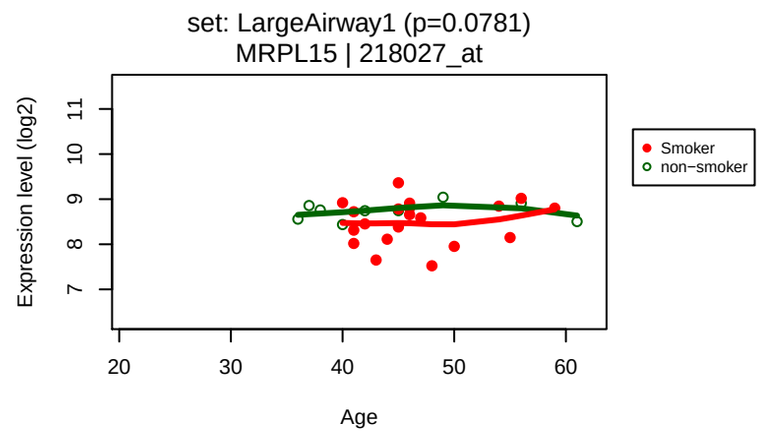

Gene expression of smokers and nonsmokers
gene: MRPL15 | 218027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0234
p=0.6457
p=0.3692
p=0.0781
p=0.2981
p=0.1985
p=0.7124
p=0.0045
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

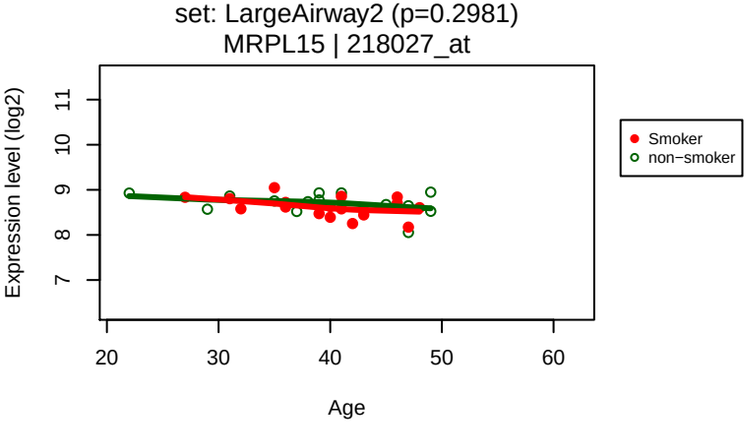

Gene expression ratio (smokers/nonsmokers)
gene: MRPL15 | 218027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0234
p=0.6457
p=0.3692
p=0.0781
p=0.2981
p=0.1985
p=0.7124
p=0.0045
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

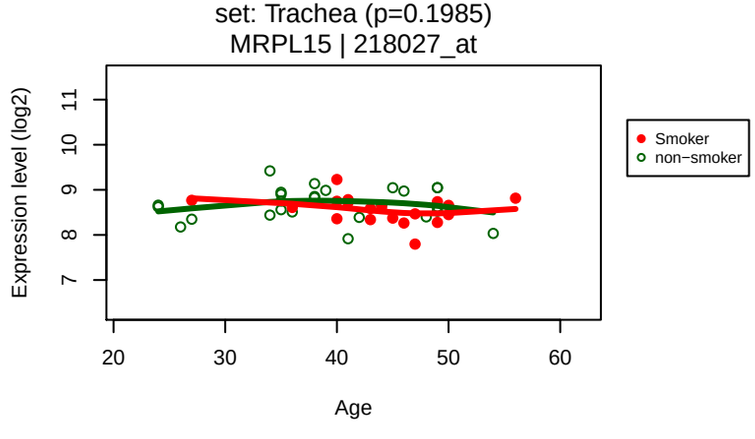

Gene expression ratio in Males
gene: MRPL15 | 218027_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7348
p=0.9575
p=0.0547
p=0.1085
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL15 | 218027_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4595
p=0.11
p=0.8458
p=0.0172
SmallAirway2
LargeAirway2
Trachea
Macrophage2

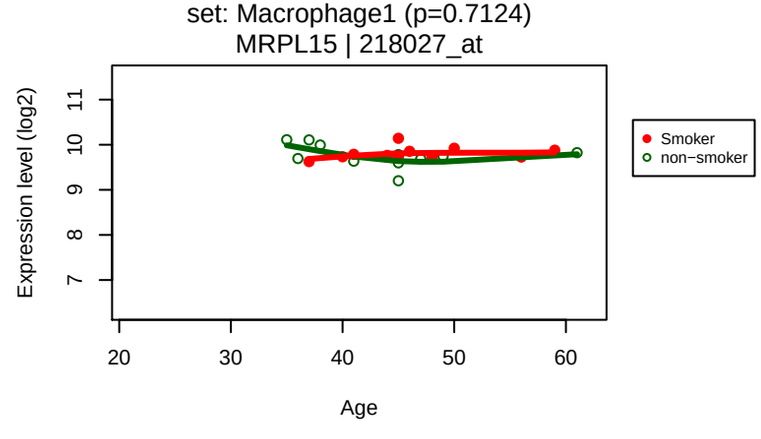

set: SmallAirway1 (p=0.0234)
MRPL15 | 218027_at
Expression level (log2)
Age
Smoker
non−smoker

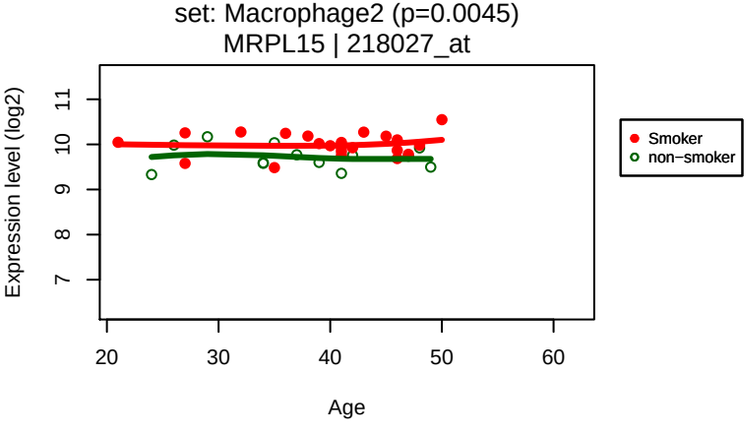

set: SmallAirway2 (p=0.6457)
MRPL15 | 218027_at
Expression level (log2)
Age
Smoker
non−smoker

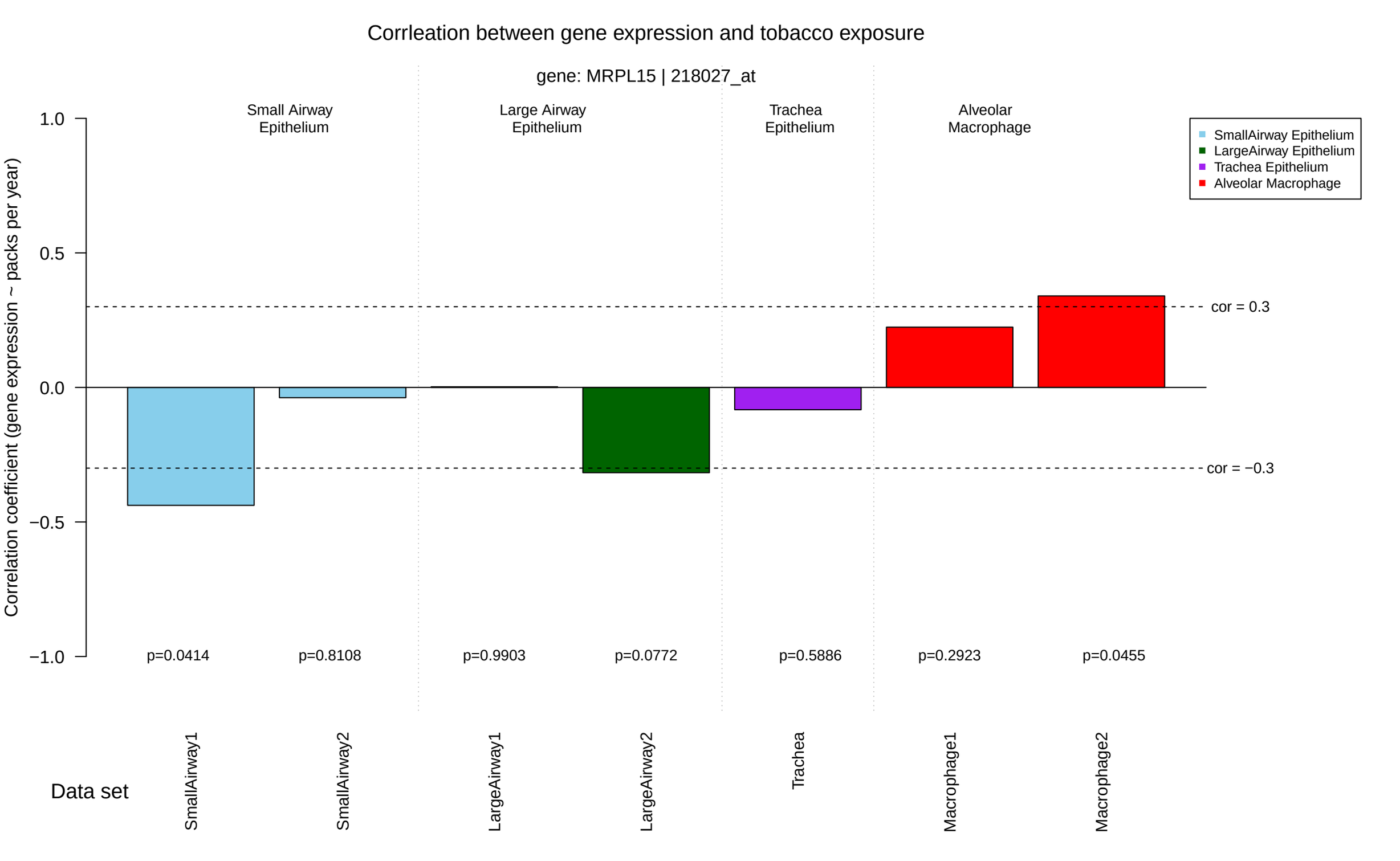

set: LargeAirway1 (p=0.0781)
MRPL15 | 218027_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2981)
MRPL15 | 218027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1985)
MRPL15 | 218027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7124)
MRPL15 | 218027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0045)
MRPL15 | 218027_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL15 | 218027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0414
p=0.8108
p=0.9903
p=0.0772
p=0.5886
p=0.2923
p=0.0455
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2